

Chinmayi Baramashetru

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Canterbury, Kent, United Kingdom

OBJECTIVE

I am a computer scientist researching in programming language theory, formal verification and language based security. My work so far has focused on specifying and proving correctness, safety, and reliability of complex distributed and cyber-physical systems, using formal methods, type systems, formal verification, and automated verification tools.

EXPERIENCE

• University of Kent

March 2025 – Present

Research Associate (Postdoc)

Canterbury, United Kingdom

- Apply **formal verification and program analysis** to large C/Fortran codebases, focusing on correctness and robustness of complex parallel infrastructure.
- Develop **formal semantics and a type system for a verifiable subset of Fortran**.
- Design **interface contracts and invariants** for inter-component communication, revealing subtle errors in data handling and control flow.
- Collaborate closely with domain scientists to translate **informal requirements** into rigorous specifications, iterating based on counterexamples and verification traces.

• University of Oslo

November 2020 – February 2025

PhD Research Fellow

Oslo, Norway

- Designed and formalised a **programming language with active object core with built-in security and privacy guarantees** using information-flow control, type systems, and policy specification.
- Defined formal semantics, type system and **proved compliance and safety properties** for distributed systems handling sensitive data.
- Implemented prototype **verification and analysis tools** (e.g. type checkers and static analyses) and evaluated them on representative case studies.

• IORA Healthcare

2019 – 2020

Data Analyst

London, United Kingdom

- Conducted reliable **data analysis and visualisations** for commercial healthcare purposes.
- Maintained and modelled **automated data collection methods** to streamline workflows.
- Manipulated and analyzed **medium- to large-scale data sets** for process optimisation.

• TATA Consultancy Services- Apple iCloud Account

2016 – 2018

Assistant System Engineer

Bangalore, India

- Automated troubleshooting of **Solaris, Linux, and Docker server operations**.
- Maintained **iCloud services**, with experience in IMAP, SMTP, DNS, and related protocols.
- Gained practical experience with **OS upgrades, patches, and configuration management**.
- Worked with **proof-points, milters, puppet, and containerized environments**.

TEACHING EXPERIENCE

- **Teaching Assistant:** Models of Concurrency (2021–2022, UiO), Developed teaching materials on concurrency models and type theory; conducted practical sessions covering core concepts with hands-on examples in Java and Rust.
- **Guest Lecturer:** Adaptive Methods for Data-Based Decision Making (2022, UiO), Delivered lectures on privacy-preserving data analysis, including anonymisation techniques, encryption methods, and secure data processing approaches.
- **Guest Lecturer:** Programming Languages (2023, UiO), Lectured on Rust programming with a focus on memory models, type systems, and associated programming paradigms.
- **Teaching Assistant:** Functional Programming (Present, University of Kent), Supported instruction in functional programming concepts and practical programming in Haskell.

EDUCATION

- **University of Oslo** Nov 2020 – Feb 2025
PhD in Computer Science Oslo, Norway
 - Thesis Title: **Data Privacy Compliance by Construction in Distributed Systems** ([Link](#))
- **City, University of London** Oct 2018 – Feb 2020
MS in Information Systems and Technology London, United Kingdom
 - Supervisor: Nyx Martin Brian ([Webpage](#))
 - Thesis Title: **Exploitation Development in Ada** to detect memory-based vulnerabilities
 - Grade: Merit
- **BIET, Karnataka** July 2012 – Oct 2016
BEng Electronics and Communication Engineering Karnataka, India
 - Grade: First Class with Distinction
 - Dean's List

PROJECTS

- **Project: Type Systems for Data Privacy Compliance** 2019 – 2021
Keywords: Rocq, Maude, Type Theory, Formal Verification [\[Publication\]](#)
 - Designed a type system to enforce data privacy compliance in active object languages used in **distributed and data-intensive systems**.
 - Implemented policy specification and enforcement to ensure security guarantees at the language level.
 - Several papers in peer-reviewed venues.
- **Project: Security Analysis — Exploit Research in Ada** Mar 2019 – Aug 2019
Keywords: C, Ada, GDB, Memory Safety, Static Analysis
 - Investigated **memory-safety vulnerabilities** (buffer overflows, heap bugs, format-string issues) in Ada and underlying C components through controlled exploit development.
 - Used **low-level debugging and instrumentation** (e.g. GDB) to study stack layouts, calling conventions, and undefined behaviour, and to understand how attacks propagate at runtime.
 - Analysed the effectiveness of compiler options and coding patterns, and recommended **safer designs and tool configurations** to harden systems against exploitation.
- **Project: GreenhouseDT – Digital Twin Exemplar** 2021 – 2022
Keywords: Self-adaptive systems, Digital twins, Knowledge representation [\[Publication\]](#)
 - Developed a **digital twin exemplar (GreenhouseDT)** to model and verify system behaviours and control strategies, supporting **simulation-driven decision making**.
 - Applied formal verification and reproducibility checks to **large-scale simulation runs**, focusing on robustness of configuration and data flows.
 - Bridged high-level system modelling with formal methods to improve transparency and trust in complex, data-intensive digital twins.
- **Project: Modelling and Verification of Coupler Behaviour in Climate Models** 2025
Keywords: Fortran, Frama-C, ACSL, Fortran/C Interoperability, Runtime Verification [\[Accepted\]](#)
 - Investigated formal verification of **coupler components** that coordinate data exchange between parallel submodels in large simulation codes.
 - Applied **Design by Contract** and ACSL specifications to detect interface-level errors in data exchange, time-stepping, and state consistency across components.
 - Developed a **hybrid verification strategy** combining static verification (Frama-C) with runtime monitoring (E-ACSL) for performance-critical C/Fortran code.

PUBLICATIONS

C=CONFERENCE/WORKSHOP, J=JOURNAL, B=BOOK / EDITED VOLUME

- [C.1] E. Kamburjan, R. Sieve, C. Prabhu Baramashetru, M. Amato, G. Barmina, E. Occhipinti, and E. Broch Johnsen. (2024). **GreenhouseDT: An exemplar for digital twins**. In *Proceedings of the 19th International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS '24)*, pp. 175–181. ACM.
- [J.1] C. Prabhu Baramashetru, P. Giannini, S. Lizeth Tapia Tarifa, and O. Owe. (2025). **A Type System for Data Privacy Compliance in Active Object Languages**. *Art, Science, and Engineering of Programming*, Vol. 10, Issue 2.
- [B.1] C. Prabhu Baramashetru, S. Lizeth Tapia Tarifa, O. Owe. (2024). **Assuring GDPR Conformance Through Language-Based Compliance**. In **Privacy and Identity Management: Sharing in a Digital World** (pp. 46–63). Springer.
- [C.2] C. Prabhu Baramashetru, S. Lizeth Tapia Tarifa, O. Owe. (2022). **A Policy Language to Capture Compliance of Data Protection Requirements**. In *Integrated Formal Methods (IFM)*, pp. 289–309. Springer.
- [C.3] C. Prabhu Baramashetru, S. Lizeth Tapia Tarifa, O. Owe. (2024). **Integrating Data Privacy Compliance in Active Object Languages**. In **Active Object Languages: Current Research Trends**, pp. 263–288. Springer.
- [C.4] C. Prabhu Baramashetru, Dominic Orchard. (2025). **Towards Modelling and Verification of Coupler Behaviour in Climate Models**. PROPL 2025 (ICFP/SPLASH workshop).

SKILLS

- **Programming Languages:** C, FORTRAN, Python, Java, Haskell, JavaScript, Go
- **Formal Verification & Analysis Tools:** Frama-C, KeY for Java, Maude, Z3, Rocq, Isabelle, Lean (beginner), SMT/SAT-based reasoning, Model-checkers
- **Other Tools:** Git, Docker, CI/CD, L^AT_EX
- **Theoretical Background:** Operational semantics, concurrency models, language-based security, distributed/cyber-physical system and PL theory
- **Version Control & Automation:** Git, scripting for build, test, and verification runs, experience maintaining small analysis and verification workflows
- **Research & Communication:** Formal modelling, proof development, experimental evaluation, scientific and technical writing, presentations for both technical and mixed audiences

HONORS AND AWARDS

- **OPLSS Fellowship, Oregon, USA** July 2022
Oregon Programming Languages Summer School
 - Awarded a competitive fellowship to attend the OPLSS summer school.
 - Recognised for research potential in programming languages and formal methods.
- **British Council 70th Anniversary Scholarship for Women in STEM** Oct 2018
British Council, UK [British Council]
 - Awarded a fully funded scholarship to support Indian women in STEM for a UK Master's degree.
 - One of the three candidates who won the scholarship at City, University of London.

COMMUNITY SERVICE

- **PC Member:** PROPL'25, FASE'22, AOL'23
- **Reviewer:** iFM'22, SEFM'21, F-IDE'21, TACAS'24, FM'24, PROPL'25
- **Volunteer:** POPL'24

PRESENTATIONS

- **Conferences:** iFM'22, NISK'21, AOL'23, IFIPSC'23
- **Workshops:** NWPT'21, ISOLA'22, NCAR'25

ADDITIONAL INFORMATION

- **Languages:** English (Native), Norwegian (Intermediate), German (Beginner), Hindi (Intermediate)
- **Citizenship:** Indian